

Lecture 5: Solving Linear Equations and Null Space

MATH203 Applied Linear Algebra

Motivation: Why We Need to Solve Equations

Recall: The Two Languages (Lecture 4)

Every subspace has **two complementary descriptions**:

Language	How it works	Example
Descriptive	List equations (constraints)	$\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$
Constructive	Provide recipe (generate members)	$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots\}$

Today's question: What is each language **good at**? What is each language **bad at**?

Each Language Has Strengths and Weaknesses

	Descriptive $\{x : Ax = 0\}$	Constructive $\text{span}\{v_1, \dots\}$
Good at	Verification (check membership)	Generation (produce members)
Bad at	Generation (find examples)	Verification (check membership)

Key insight: Solving equations helps both languages do what they're bad at!

Let's see two concrete examples using Shinchon's coffee shop...

Example Setup: Shinchin's Coffee Shop Menu

Shinchan has three drinks on his menu:

	 milk	 coffee	 tea
 tea leaves	0	0	2
 lemon	0	0	1
 beans	0	2	4
 milk	1	0	1

Matrix form:

$$A = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 4 \\ 1 & 0 & 1 \end{pmatrix}$$

Case 1: Column Space (Naturally Constructive)

The column space is **naturally constructive**:

$$\text{Col}(A) = \text{span} \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} \right\}$$

These are the ingredient lists for , , .

In coffee shop language: $\text{Col}(A)$ = all possible ingredient combinations we can produce from our menu.

Case 1: Generation is Easy

Generation (easy): Want a vector in $\text{Col}(A)$?

Just pick a combination of drinks!

$$2 \text{ 🥛 } + 3 \text{ ☕ } = 2 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 6 \\ 2 \end{pmatrix}$$

In coffee shop language: "Make 2 glasses of milk and 3 cups of coffee" produces an ingredient combination in $\text{Col}(A)$.

Generation = **making products**. Easy with constructive description!

Case 1: Verification is Hard

Verification (hard): A customer brings an ingredient list $\mathbf{b} = \begin{pmatrix} 4 \\ 2 \\ 10 \\ 3 \end{pmatrix}$.

Question in coffee shop language: Can we make a drink combination that uses exactly these ingredients?

This is asking: Is $\mathbf{b} \in \text{Col}(A)$?

Can't tell by looking! Need to **solve**: $A\mathbf{x} = \mathbf{b}$

- If solution exists $\Rightarrow$ Yes, we can make it! ($\mathbf{b} \in \text{Col}(A)$)
- If no solution $\Rightarrow$ No, impossible with our menu ($\mathbf{b} \notin \text{Col}(A)$)

Solving equations gives constructive description the power to verify!

Case 2: Null Space (Naturally Descriptive)

The null space is **naturally descriptive**:

$$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^3 : A\mathbf{x} = \mathbf{0}\}$$

In coffee shop language: $\text{Null}(A)$ = records of relationships between drinks.

For example: If  can be made from $2 \times$ , then the relationship is recorded as:

$$\begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} : \quad 2 \text{ coffees can make 1 tea}$$

Geometric meaning: $\text{Null}(A)$ stores all relationships between the columns.

Case 2: Verification is Easy (Coffee Shop Example)

Verification (easy): Someone claims "🍵 can be made from 1 🥛 and 1 ☕."

Is this relationship valid? Check if $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \in \text{Null}(A)$:

Compute:

$$\begin{aligned} 1 \text{ 🥛} + 1 \text{ ☕} - 1 \text{ 🍵} &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} -2 \\ -1 \\ -2 \\ 0 \end{pmatrix} \neq \mathbf{0} \end{aligned}$$

Case 2: Generation is Hard

Generation (hard): Looking at the menu, can you find the relationships?

			
	0	0	2
	0	0	1
	0	2	4
	1	0	1

Question in coffee shop language: Which drinks can be made from others? What are the relationships?

Hard to see by staring! Need to **solve** $Ax = 0$.

Case 2: Solving Reveals the Relationship

After solving $A\mathbf{x} = \mathbf{0}$ (we'll learn how soon):

$$\text{Null}(A) = \text{span} \left\{ \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} \right\}$$

What does this tell us in coffee shop language?

$$\begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} : \text{ 🍵 can be made from 2 🍵 }$$

Aha! This is the **relationship**: 1 cup of 🍵 uses exactly the same ingredients as 2 cups of 🍵.

Solving equations gives descriptive description the power to

Summary: The Power of Solving Equations

Subspace	Weakness	Solving provides
$\text{Col}(A)$ (constructive)	Verification "Can we make this?"	Solve $A\mathbf{x} = \mathbf{b}$ to check if possible
$\text{Null}(A)$ (descriptive)	Generation "What relationships?"	Solve $A\mathbf{x} = \mathbf{0}$ to find relationships

Today's focus: Develop a **cross-filling method** to solve $A\mathbf{x} = \mathbf{b}$, which:

- Finds all relationships between products (basis for $\text{Null}(A)$)
- Checks if a product is possible ($\mathbf{b} \in \text{Col}(A)$?)
- Determines uniqueness (Is there only one way?)

Null Space: The Relationship Catalog

Definition: Null Space

Definition (Null Space (Kernel))

For a matrix $A \in \mathbb{R}^{m \times n}$, the **null space** is:

$$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

Meaning: $\text{Null}(A)$ records all **relationships between the columns** of A .

Also called the **kernel**, denoted $\ker(A)$.

In coffee shop language: Drink combinations that reveal "which drinks can be made from others."

Immediate Example: Is This a Valid Relationship?

From Shinnan's menu (A from before), someone claims:

" can be made from 2 "

This means: $\begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$ should be in $\text{Null}(A)$.

Let's verify:

$$A \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = 0 \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + 2 \cdot \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} - 1 \cdot \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 0 \\ -1 \end{pmatrix} \neq \mathbf{0}$$

Result: This is NOT a valid relationship! (Actually  $\neq 2 \times$ )

What Would Be a Valid Relationship?

Let's try another: someone claims "  can be made from 2  plus 1  "

This means: $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ might be in $\text{Null}(A)$.

$$A \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 0 \\ 0 \end{pmatrix} \neq \mathbf{0}$$

Still not valid!

How do we **systematically find** valid relationships? We need to solve $A\mathbf{x} = \mathbf{0}$!

Key Insight: Null Space Records Relationships

Important:

- Each vector $\mathbf{x} \in \text{Null}(A)$ records a relationship
- $\text{Null}(A) = \text{"relationship catalog"}$ between columns

Size of the catalog:

- $\text{Null}(A) = \{\mathbf{0}\} \rightarrow$ no relationships $\rightarrow$ columns **independent**
- $\dim(\text{Null}(A)) = k \rightarrow$ there are k independent relationships

Null Space is a Subspace

Proposition

$\text{Null}(A)$ is a subspace of $\mathbb{R}^n$.

Proof: Verify the three axioms:

1. $\mathbf{0} \in \text{Null}(A)$ since $A\mathbf{0} = \mathbf{0}$ ✓
2. If $A\mathbf{x}_1 = \mathbf{0}$ and $A\mathbf{x}_2 = \mathbf{0}$, then

$$A(\mathbf{x}_1 + \mathbf{x}_2) = \mathbf{0}$$

✓

3. If $A\mathbf{x} = \mathbf{0}$, then $A(c\mathbf{x}) = \mathbf{0}$ ✓

The Problem: We Need Constructive Description

Descriptive: $\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$

- **Verification** (easy): Is $\mathbf{v}$ a valid relationship? Compute $A\mathbf{v}$
- **Generation** (hard): Find all relationships? Can't guess!

Goal: Convert to **constructive**:

$$\text{Null}(A) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$$

Method: Solve $A\mathbf{x} = \mathbf{0}$ using cross-filling!

But first — to solve $A\mathbf{x} = \mathbf{b}$ in general, we need augmented matrices...

From $Ax = b$ to Augmented Matrix

Motivation: Why Augmented Matrix?

We want to solve: $A\mathbf{x} = \mathbf{b}$

Column view:

$$A\mathbf{x} = \begin{pmatrix} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

We're looking for **coefficients** $x_1, \dots, x_n$ that combine columns to get $\mathbf{b}$.

Key idea: Add $\mathbf{b}$ as an extra column...

The Augmented Matrix

Form the **augmented matrix**:

$$(A \mid \mathbf{b}) = \left(\begin{array}{c|c|c|c|c} | & | & & | & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \\ | & | & & | & | \end{array} \right)$$

This is an $m \times (n + 1)$ matrix with $n + 1$ columns.

Now: What happens if we take a linear combination with coefficients

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \\ -1 \end{pmatrix} ?$$

The Key Connection

Take $(A \mid \mathbf{b})$ and multiply by $\begin{pmatrix} x_1 \\ \vdots \\ x_n \\ -1 \end{pmatrix}$:

$$(A \mid \mathbf{b}) \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ -1 \end{pmatrix} = x_1 \mathbf{a}_1 + \cdots + x_n \mathbf{a}_n + (-1)\mathbf{b}$$

This equals $\mathbf{0}$ if and only if:

$$x_1 \mathbf{a}_1 + \cdots + x_n \mathbf{a}_n = \mathbf{b}$$

which is exactly $A\mathbf{x} = \mathbf{b}$!

Why This Matters

Crucial observation:

$A\mathbf{x} = \mathbf{b}$ has solution $\iff \begin{pmatrix} \mathbf{x} \\ -1 \end{pmatrix}$ records a relationship in $(A \mid \mathbf{b})$

But we won't compute $\text{Null}(A \mid \mathbf{b})$ directly. Instead:

We'll use **cross-filling** on $(A \mid \mathbf{b})$ to extract simplified equations, then solve backwards!

The Cross-Filling Method for Solving $Ax = b$

Recall: Cross-Filling Decomposition (Lecture 3)

Any matrix can be decomposed via cross-filling:

$$A = R_1 + R_2 + \cdots + R_r$$

where each R_i is a **rank-one matrix**.

Key property: All rows of a rank-one matrix are scalar multiples of each other.

Example:

$$R = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{pmatrix}$$

Apply Cross-Filling to Augmented Matrix

For $(A \mid \mathbf{b})$, we cross-fill:

$$(A \mid \mathbf{b}) = (R_1 \mid \mathbf{b}_1) + (R_2 \mid \mathbf{b}_2) + \cdots + (R_r \mid \mathbf{b}_r)$$

Each piece $(R_i \mid \mathbf{b}_i)$ is a **rank-one augmented matrix** (includes both A and $\mathbf{b}$ parts).

Key observation: Since $(R_i \mid \mathbf{b}_i)$ is rank-one, all its rows are proportional — they represent the **same equation!**

One rank-one piece = one independent equation.

Why Are These Equations Valid?

Question: Why is solving these r equations equivalent to the original m equations?

Answer: Cross-filling gives $(A \mid \mathbf{b}) = UV$ where:

- U is $m \times r$ with **linearly independent columns**
- V is $r \times (n + 1)$

Since U 's columns are independent:

$$UV\mathbf{z} = \mathbf{0} \iff UV\mathbf{z} = U\mathbf{0} \iff V\mathbf{z} = \mathbf{0}$$

(Left multiplication by independent columns can be cancelled.)

What is V Explicitly?

$$V = \begin{pmatrix} \text{— row from } (R_1 \mid \mathbf{b}_1) \text{ —} \\ \text{— row from } (R_2 \mid \mathbf{b}_2) \text{ —} \\ \vdots \\ \text{— row from } (R_r \mid \mathbf{b}_r) \text{ —} \end{pmatrix}$$

V is an $r \times (n + 1)$ **augmented matrix**, where:

- Each row is one simplified equation
- The last column corresponds to the right-hand side

In practice: We'll see V explicitly in our example!

Observation: Variables Decrease

As cross-filling proceeds, each piece has fewer active variables:

- $(R_1 \mid \mathbf{b}_1)$: involves **all n variables**
- $(R_2 \mid \mathbf{b}_2)$: involves **fewer variables** (at least column j_1 eliminated)
- $(R_3 \mid \mathbf{b}_3)$: involves **even fewer variables**
- $\vdots$
- $(R_r \mid \mathbf{b}_r)$: involves **fewest variables**

Key: Each successive piece has **at least one fewer variable** (possibly more, depending on which columns are eliminated).

Strategy: Start from the end (fewest variables) and work backwards!

Concrete Example: Step-by-Step Visual Cross-Filling

Example Setup

Solve:

$$A\mathbf{x} = \mathbf{b}$$

where

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 2 \\ 3 & 6 & 1 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 11 \\ 16 \end{pmatrix}$$

Form augmented matrix $(A \mid \mathbf{b})$ and cross-fill visually!

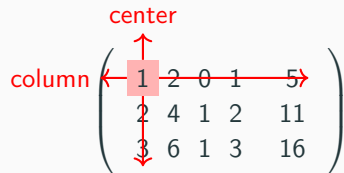
Step 1: Initial Augmented Matrix

$$\left(\begin{array}{ccccc} 1 & 2 & 0 & 1 & 5 \\ 2 & 4 & 1 & 2 & 11 \\ 3 & 6 & 1 & 3 & 16 \end{array} \right)$$

Goal: Use cross-filling to decompose into rank-one pieces.

Strategy: Choose pivots to eliminate columns one by one.

Step 2: Choose First Pivot at (1,1)



The diagram shows a matrix with a pivot element at (1,1) highlighted in red. Red arrows point from the pivot to the rest of the matrix, indicating row and column operations. The word "center" is written above the pivot, and "column" is written to the left of the pivot.

$$\text{center} \quad \text{column} \quad \left(\begin{array}{ccccc} 1 & 2 & 0 & 1 & 5 \\ 2 & 4 & 1 & 2 & 11 \\ 3 & 6 & 1 & 3 & 16 \end{array} \right)$$

Pivot = entry $(1,1) = 1$ (highlighted red).

Extract: Column $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \text{Row } (1 \quad 2 \quad 0 \quad 1 \quad | \quad 5)$

Step 3: First Rank-One Piece ($R_1 \mid \mathbf{b}_1$)

$$(R_1 \mid \mathbf{b}_1) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 5 \end{array} \right)$$

Expanding:

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 5 \\ 2 & 4 & 0 & 2 & 10 \\ 3 & 6 & 0 & 3 & 15 \end{array} \right) \leftarrow (R_1 \mid \mathbf{b}_1)$$

Equation from ($R_1 \mid \mathbf{b}_1$) (read any row, e.g., row 1):

$$x_1 + 2x_2 + 0x_3 + 1x_4 = 5$$

Step 4: Compute Remainder

Before: $(A \mid \mathbf{b})$

$$\left(\begin{array}{ccccc} 1 & 2 & 0 & 1 & 5 \\ 2 & 4 & 1 & 2 & 11 \\ 3 & 6 & 1 & 3 & 16 \end{array} \right) - \left(\begin{array}{ccccc} 1 & 2 & 0 & 1 & 5 \\ 2 & 4 & 0 & 2 & 10 \\ 3 & 6 & 0 & 3 & 15 \end{array} \right)$$

After: Remainder

$$\left(\begin{array}{ccccc} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right)$$

Column 1 eliminated! Next pivot at $(2, 3)$ (blue).

Step 5: Choose Second Pivot at (2,3)

$$\left(\begin{array}{ccccc} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right)$$

Pivot = entry $(2, 3) = 1$ (blue).

Extract: Column $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \times \text{Row } (0 \ 0 \ 1 \ 0 \mid 1)$

Step 6: Second Rank-One Piece ($R_2 \mid \mathbf{b}_2$)

$$(R_2 \mid \mathbf{b}_2) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \left(\begin{array}{cccc|c} 0 & 0 & 1 & 0 & 1 \end{array} \right)$$

Expanding:

$$\left(\begin{array}{cccc|c} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right) \leftarrow (R_2 \mid \mathbf{b}_2)$$

Equation from ($R_2 \mid \mathbf{b}_2$) (read any non-zero row):

$$x_3 = 1$$

Step 7: Final Remainder is Zero

After subtracting $(R_2 \mid \mathbf{b}_2)$:

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Cross-filling complete!

$$(A \mid \mathbf{b}) = (R_1 \mid \mathbf{b}_1) + (R_2 \mid \mathbf{b}_2)$$

Step 8: The Matrix V

Recall: $(A \mid \mathbf{b}) = UV$ where rows of V come from each $(R_i \mid \mathbf{b}_i)$.

Explicitly:

$$V = \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 5 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right)$$

- Row 1 from $(R_1 \mid \mathbf{b}_1)$: $x_1 + 2x_2 + 0x_3 + x_4 = 5$
- Row 2 from $(R_2 \mid \mathbf{b}_2)$: $0x_1 + 0x_2 + x_3 + 0x_4 = 1$

Solving $(A \mid \mathbf{b})$ is equivalent to solving V !

Step 9: How to Identify Pivot and Free Variables

From $(R_2 \mid \mathbf{b}_2)$ to $(R_1 \mid \mathbf{b}_1)$, we went from 1 active variable to 4 active variables.

Which new variables appeared? x_1, x_2, x_4 (three new variables)

Which one did we choose as pivot in $(R_1 \mid \mathbf{b}_1)$? x_1 (at position (1,1))

Therefore:

- $x_1 =$ **pivot variable** (we chose it)
- $x_2, x_4 =$ **free variables** (appeared but not chosen)

General rule: From $(R_{i+1} \mid \mathbf{b}_{i+1})$ to $(R_i \mid \mathbf{b}_i)$, new variables appear. The one we chose as pivot = pivot variable. Others = free variables.

Step 10: Summary of Variables

Pivot columns: 1, 3

- **Pivot variables:** x_1, x_3 (determined by equations)

Non-pivot columns: 2, 4

- **Free variables:** x_2, x_4 (can choose any value)

Existence check: No equation of form $0 = c$ ($c \neq 0$) found.

⇒ **Solution exists!**

Step 11: Backward Solving

From $(R_2 \mid \mathbf{b}_2)$ (simplest equation):

$$x_3 = 1$$

From $(R_1 \mid \mathbf{b}_1)$:

$$x_1 + 2x_2 + 0x_3 + x_4 = 5$$

Substituting $x_3 = 1$:

$$x_1 + 2x_2 + 0(1) + x_4 = 5$$

$$x_1 + 2x_2 + x_4 = 5$$

Solve for pivot variable x_1 :

Step 12: Write General Solution

Let free variables $x_2 = s$ and $x_4 = t$ (parameters).

General solution:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 5 - 2s - t \\ s \\ 1 \\ t \end{pmatrix}$$

Decompose:

$$= \begin{pmatrix} 5 \\ 0 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Step 13: Interpret the Solution

$$\mathbf{x} = \underbrace{\begin{pmatrix} 5 \\ 0 \\ 1 \\ 0 \end{pmatrix}}_{\text{Particular solution } \mathbf{x}_p} + s \underbrace{\begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}}_{\text{Null vector } \mathbf{v}_1} + t \underbrace{\begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}}_{\text{Null vector } \mathbf{v}_2}$$

Particular solution ($s = t = 0$): $\mathbf{x}_p = \begin{pmatrix} 5 \\ 0 \\ 1 \\ 0 \end{pmatrix}$

Null space basis:

$$\text{Null}(A) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Applications: Existence, Uniqueness, Null Space

Application 1: Finding Null Space Basis

Proposition

To find a basis for $\text{Null}(A)$:

1. Solve $A\mathbf{x} = \mathbf{0}$ using cross-filling
2. Identify free variables
3. For each free variable, set it = 1, others = 0
4. Solve for pivot variables

Result: $\text{Null}(A) = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$

Descriptive $\rightarrow$ Constructive conversion complete!

Application 1: Concrete Example

From our example, free variables are x_2, x_4 .

For $\mathbf{v}_1$: Set $x_2 = 1, x_4 = 0$

$$x_3 = 1 \quad \text{wait, this is for } A\mathbf{x} = \mathbf{b}!$$

Correction: For null space, solve $A\mathbf{x} = \mathbf{0}$ (not $\mathbf{b}$)!

From $(R_1 \mid \mathbf{0})$: $x_1 + 2x_2 + x_4 = 0$, so $x_1 = -2x_2 - x_4$

From $(R_2 \mid \mathbf{0})$: $x_3 = 0$

$$\text{Set } x_2 = 1, x_4 = 0: \mathbf{v}_1 = \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{Set } x_2 = 0, x_4 = 1: \mathbf{v}_2 = \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Application 2: Existence Criterion — The Key Insight

Question: When does $A\mathbf{x} = \mathbf{b}$ have a solution?

Strategy: Cross-fill $(A \mid \mathbf{b})$ using pivots from A only.

- Use $r = \text{rank}(A)$ pivots to eliminate A part
- After r steps, all pivot rows are cleared
- Then check: **What's left in the non-pivot rows?**

Let's see this visually with colored rows. . .

Case 1: All Equations Effective (Solution Exists)

After using r pivots from A :

$$\begin{array}{l} \text{pivot row 1} \\ \text{pivot row 2} \\ \text{non-pivot row} \end{array} \left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Observation:

- Pivot rows (colored): cleared by design
- Non-pivot rows: **passively became 0** (including **b** part)

Conclusion: No equation of form $0 = c$ ($c \neq 0$).

All equations are **effective** (not contradictory).

Solution exists!

Case 2: Contradiction Appears (No Solution)

After using r pivots from A :

$$\begin{array}{l} \text{pivot row 1} \\ \text{pivot row 2} \\ \text{non-pivot row} \end{array} \left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & c \end{array} \right) c \neq 0$$

Observation:

- Pivot rows: cleared
- Non-pivot row: A part is 0, but $\mathbf{b}$ part $\neq 0$

This gives equation: $0 = c$ where $c \neq 0$ — **contradiction!**

No solution!

Existence Theorem: Correct Logic

Theorem (Existence of Solutions)

$A\mathbf{x} = \mathbf{b}$ has a solution if and only if:

$$\text{rank}(A \mid \mathbf{b}) = \text{rank}(A)$$

Correct logic:

- The cross-filling algorithm **always proceeds**
- If $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A) \Rightarrow$ no contradiction $\Rightarrow$ solution exists
- If $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A) + 1 \Rightarrow$ contradiction $0 = c$ appears $\Rightarrow$ **any $\mathbf{x}$ violates this equation** $\Rightarrow$ no solution

Geometric Interpretation

$$\mathbf{b} \in \text{Col}(A) \iff \text{rank}(A \mid \mathbf{b}) = \text{rank}(A)$$

Adding $\mathbf{b}$ as a column doesn't increase rank $\iff \mathbf{b}$ is already in span of A 's columns.

Coffee shop: Customer's ingredient list is achievable $\iff$ it's in our column space.

Application 3: Uniqueness Criterion

Theorem (Uniqueness of Solutions)

If $A\mathbf{x} = \mathbf{b}$ has a solution, it is **unique** if and only if:

$$\text{Null}(A) = \{\mathbf{0}\}$$

Equivalently:

- No free variables
- All columns are pivot columns
- $\text{rank}(A) = n$ (number of columns)

Proof idea: If $\mathbf{x}_1, \mathbf{x}_2$ are both solutions, then $\mathbf{x}_1 - \mathbf{x}_2 \in \text{Null}(A)$.

Application 4: Structure of Solution Set

Theorem (General Solution Structure)

If $A\mathbf{x} = \mathbf{b}$ has solutions, **all solutions** have the form:

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$$

where:

- $\mathbf{x}_p$ is any **particular solution**
- $\mathbf{x}_n \in \text{Null}(A)$ is arbitrary

Geometric picture: Solution set is $\mathbf{x}_p + \text{Null}(A)$ (affine subspace).

Dimension Formula: Rank-Nullity Theorem

Counting Pivot and Free Variables

From cross-filling an $m \times n$ matrix A :

- Number of pivot columns = $\text{rank}(A)$
- Number of free variables = $\dim(\text{Null}(A))$
- Total columns = n

Every column is either pivot or non-pivot:

$$\text{rank}(A) + \dim(\text{Null}(A)) = n$$

This is the **Rank-Nullity Theorem**.

Visual Example: Rank-Nullity

From our example:

$$\begin{array}{c} \text{pivot} \quad \text{free} \quad \text{pivot} \quad \text{free} \\ \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 0 & 1 \\ \hline 2 & 4 & 1 & 2 \\ \hline 3 & 6 & 1 & 3 \\ \hline \end{array} \right) \end{array}$$

Pivot columns (rank = 2)

Non-pivot columns (nullity = 2)

$$\underbrace{2}_{\text{rank}} + \underbrace{2}_{\text{nullity}} = \underbrace{4}_n$$

Rank-Nullity Theorem

Theorem (Rank-Nullity Theorem)

For any $m \times n$ matrix A :

$$\text{rank}(A) + \dim(\text{Null}(A)) = n$$

Proof idea: Every column is either:

- A pivot column (contributes to rank), or
- A non-pivot column (contributes one basis vector to null space)

These counts sum to n .

Preview: Null Space and Row Space Orthogonality

Natural Orthogonality

Theorem

$\text{Null}(A) \perp \text{Row}(A)$

Every vector in $\text{Null}(A)$ is orthogonal to every vector in $\text{Row}(A)$.

Proof: Let $\mathbf{x} \in \text{Null}(A)$ and $\mathbf{r}_i = i\text{-th row}$.

By definition: $A\mathbf{x} = \mathbf{0}$ means $\mathbf{r}_i \cdot \mathbf{x} = 0$ for all i .

Coffee Shop Interpretation

If $A = \text{"drinks"} \rightarrow \text{ingredients}$:

- **Row space:** All ingredient combinations from drinks
- **Null space:** Relationships between drinks (which can be made from others)

Perpendicular: A relationship (which drinks are equivalent) has nothing in common with actual ingredient requirements!

Next lecture: Four fundamental subspaces and complete orthogonal structure.

Summary

What We Achieved Today

1. Null Space:

- Descriptive: $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$
- Records relationships between columns
- Conversion to constructive requires solving

2. Cross-Filling Method:

- Each $(R_i \mid \mathbf{b}_i) =$ one equation
- Matrix V explicitly shows simplified equations
- Variables decrease (at least one per step) $\rightarrow$ backward solving

What We Achieved Today (continued)

3. Existence: $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A) \iff$ no contradiction

- Visual: pivot rows cleared, non-pivot rows passive
- If non-pivot row has $0 = c$ ($c \neq 0$) $\rightarrow$ no solution

4. Uniqueness: $\text{Null}(A) = \{\mathbf{0}\} \iff$ unique solution

5. Rank-Nullity: $\text{rank}(A) + \text{nullity} = n$ (visual: colored columns)

6. Orthogonality: $\text{Null}(A) \perp \text{Row}(A)$

Next lecture: Four fundamental subspaces!